

GEOSCIENCES

Exceptional preservation of soft tissue in a new specimen of *Eoconfuciusornis* and its biological implications

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ABSTRACT

We report on an exceptional specimen of *Eoconfuciusornis* preserving rare soft-tissue traces of the ovary and wing. Ovarian follicles preserve a greater hierarchy than observed in *Jeholornis* and enantiornithines, suggesting confuciusornithiforms evolved higher rates of yolk deposition in parallel with the neornithine lineage. The preserved soft tissues of the wing indicate the presence of a propatagium and postpatagium, whereas an alular patagium is absent. Preserved remnants of the internal support network of the propatagium bear remarkable similarity to that of living birds. Soft tissue suggests the confuciusornithiform propatagium could maintain a cambered profile and generate lift. The feathers of the wing preserve remnants of their original patterning; however, this is not strongly reflected by observable differences under scanning electron microscopy (SEM). The tail plumage lacks elongate rectrices, suggesting that the earliest known confuciusornithiforms were sexually dimorphic in their plumage.

Keywords: Huajiyang, Jehol Biota, Aves, Confuciusornithiformes, propatagium, feathers

INTRODUCTION

Thousands of complete and articulated specimens and half the known Mesozoic diversity of fossil birds have been uncovered from the Lower Cretaceous volcano-lacustrine ‘Jehol’ strata in north-eastern China. These deposits provide a rare taphonomic window into the biology of early birds through the preservation of what are otherwise typically rare soft-tissue remnants (i.e. claw sheaths, feather remains, etc.) [1,2]. Faint impressions of feathers are common but their detailed structure is more elusive [3]. However, studies utilizing scanning electron microscopy (SEM) have had enormous success in identifying important new morphological data, such as the melanosomes primarily responsible for feather colour [4–7]. Although other types of preserved soft tissue (i.e. cartilage) are much rarer [8], recent discoveries such as the preservation of epidermal and dermal-tissue remnants and collagen fibres in an enantiornithine wing (MCCMLH31444) from the Las Hoyas Konservat-Lagerstätte [9] highlight the strong potential for soft-tissue preservation in certain depositional environments [10]. Such

traces can be highly informative; MCCMLH31444 reveals the presence of essentially modern wing patagia and an advanced degree of follicular control in the feathers of enantiornithines [10].

The soft-tissue impressions of mature ovarian follicles have recently been reported in several specimens from the Jehol Lagerstätte, revealing early stages in the evolution of the specialized modern avian reproductive system [11]. Although a controversial interpretation, no plausible alternative has been provided [12,13]. Most specimens are referable to Enantiornithes [14], the dominant clade of Cretaceous birds, although one specimen is referable to the long bony-tailed bird *Jeholornis*. In all specimens, a single cluster of follicles is present on the left side of the body, indicating that even stem birds were advanced in the fact that they had already lost the function of the right ovary [11]. Follicular morphology in the Jehol basal pygostylians *Sapeornis* and *Confuciusornis* was previously unknown.

The earliest representatives of the Jehol Biota occur in the 130.7 Ma *Propteryx*-horizon of the Huajiyang Formation [15], the second oldest avian-bearing deposits in the world, younger only than the

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Solnhöfen Limestones that produce *Archaeopteryx* [16]. During this early stage, diversity was low and only five bird taxa have been described: three enantiornithines [17], an ornithuromorph [18] and the holotype of *Eoconfuciusornis zhengi* [19], the earliest representative of the most prolific clade in the Jehol Biota, the Confuciusornithiformes. Thousands of specimens of the more derived *Confuciusornis sanctus* have been collected from younger Jehol deposits of the Yixian and Jiufotang formations [2]. *Eoconfuciusornis* is the oldest and most primitive bird with a beak [19]. Members of the Confuciusornithiformes are characterized by enlarged premaxillae, robust forelimbs with a massive humeral deltopectoral crest and enlarged manual claws—features all developed to a lesser degree in the basal *Eoconfuciusornis*.

Here we describe a second specimen of this important basal taxon (Figs 1 and 2; see Supplementary Data, available at NSR online, for taxonomic affinity and full description) with exceptional partial soft-tissue preservation of the wings, ovary and plumage. The soft tissue of the wing records an intermediate condition between that of long bony-tailed birds and ornithothoracines, whereas the ovarian follicles show a trend towards rapid yolk deposition that parallels that of neornithines.

DESCRIPTION

Ovarian follicles

The ovarian follicles are preserved *in situ*, just ventral to the thoracic vertebrae and cranial to the pelvis (Figs 1 and 2 and Supplementary Fig. 1, available as Supplementary Data at NSR online), consistent with the position of the ovary in living birds (stomach contents are more ventrally located in the abdomen). The preservation of ovarian traces is inferred to be facilitated by the multi-layered follicular wall, which becomes thicker as the follicle matures [20]. This is consistent with the fact no non-vitellogenic follicles are preserved in any known specimen. Thus, both the preservation and relatively large size of the follicles suggest yolk deposition was not in its early stages. The follicles are preserved as slightly three-dimensional voids. The ventral surface of the void bears a circular micro-pattern visible in relief under SEM (Fig. 2B and E; see Supplementary Fig. 2, available as Supplementary Data at NSR online). This pattern has not been observed in any known seed, but has been previously documented in the preserved feathers of the holotype of *Sinosauropteryx* [21]. It is unclear whether the circles are concentric or spiralling (see Supplementary Fig. 1B, available as Supplementary Data at NSR online). This pattern is inferred to be the result of

authigenic mineral growth [22], and not a natural morphology of the follicles. The circles forming the micro-pattern vary in diameter (0.2–1.0 mm across), indicating that nucleation did not begin simultaneously. Approximately 10–12 overlapping follicles are preserved. The follicles range in diameter from 3.5 to 5.8 mm (see Supplementary Fig. 1A, available as Supplementary Data at NSR online).

Soft tissue of the wing

The remains of epidermal and dermal connective tissue is preserved as a light brownish residue along the cranial margin of the propatagium and along the caudal margin of both ulnae and the hand, representing the postpatagium (Figs 1–3). The soft tissue does not fluoresce when stimulated with a 408-nm violet laser (see Supplementary Fig. 3, available as Supplementary Data at NSR online). As in MCCMLH31444, the postpatagium is wider along the hand than along the ulnae and extends up to the major digit ungual [9]. The calami of three primary rachises are visible inserting in the postpatagium on the right hand (Fig. 3 and Supplementary Fig. 4b, available as Supplementary Data at NSR online). Faint dermal traces surround the alular digit and connect the base of this digit to the major metacarpal, showing that the alular digit was free and no alular patagium was present (Fig. 3).

The cranial margin of the propatagium is delimited by feathers oriented nearly perpendicular to the dorsal margin of the humerus (Fig. 3). The dermal tissue of the propatagium is only preserved along its cranial margin, obscuring the position of the propatagial ligament. In living birds, this ligament extends from the deltopectoral crest (and/or the tendons or fascia of the m. pectoralis or m. deltoideus major), attaching distally to the joint capsule covering the wrist, the extensor process of the alular metacarpal (absent in non-ornithuromorph birds) and the alular digit [23,24] (Fig. 3). In STM7-144, the right hand is preserved slightly disarticulated from the antebrachium, which may suggest the absence of a strong ligamental attachment.

At its internal edge, the brownish layer of dermal tissue can be observed overlapping a few of the banded structures that represent part of the internal soft tissue of the propatagium, preserved in the area along the dorsal margin of the humerus, extending out as far as the proximal third of the antebrachium (Figs 2C and 3). Unlike the short body feathers, which, when preserved, have an uneven, delicate and wispy appearance, the organic remains of the internal structure of the propatagium consist of solid, even, dark bands arranged in parallel

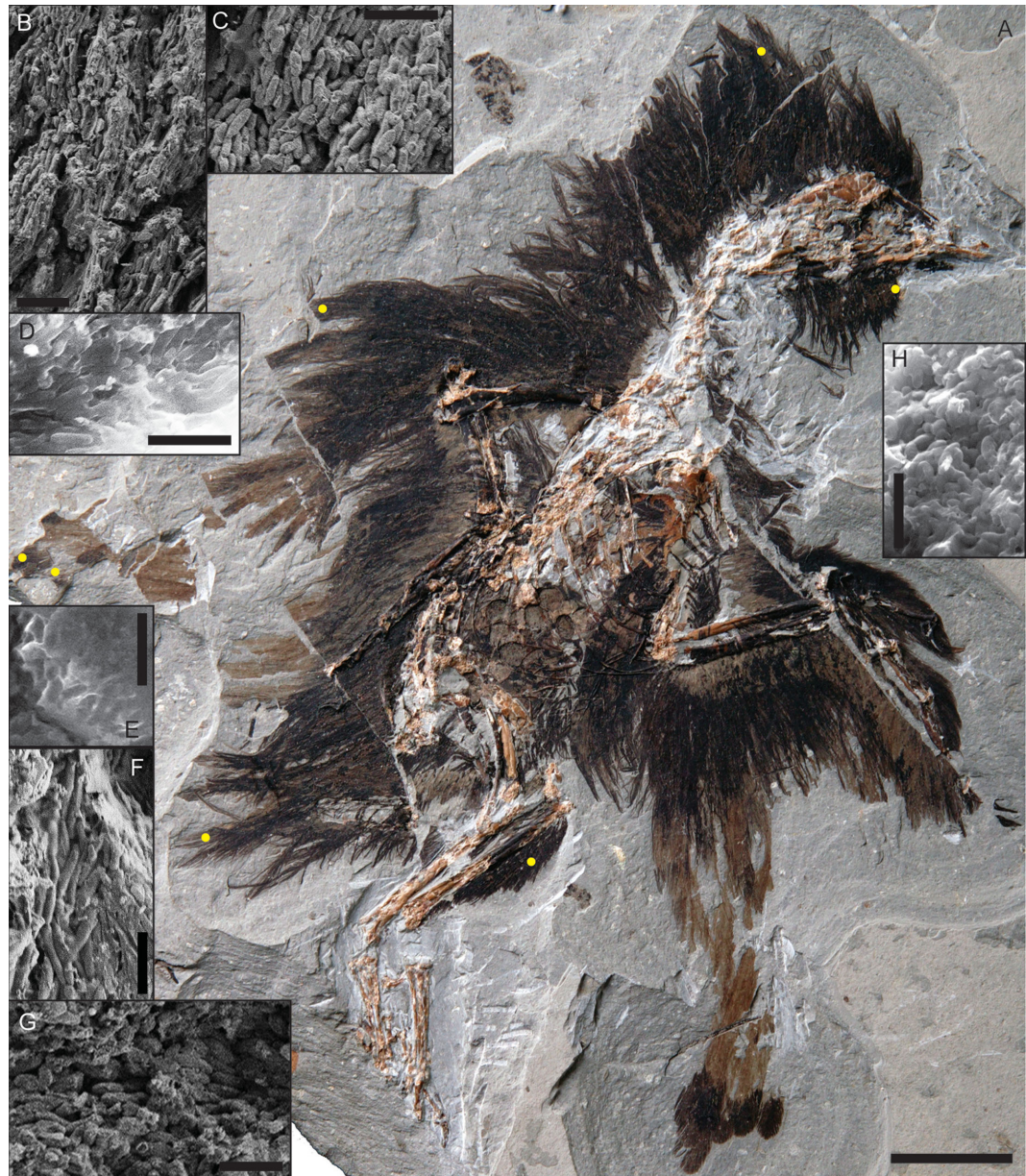


Figure 1. (A) Photograph of *Eoconfuciusornis* indet. STM7-144, preserved in right lateral view; scale bar equals 20 mm. Inset SEM images (all scale bars equal 2 μ m); (B) left-wing coverts (sample 2_2) preserving black eumelanosomes; (C) coronal feathers (sample 2_1) preserving grey eumelanosomes; (D) dark spot of secondaries (sample A) preserving black eumelanosomes; (E) light part of secondaries (sample C) preserving grey mouldic eumelanosomes; (F) tail feathers (sample 2_3) preserving black eumelanosomes; (G) crural feathers (sample 2_4) preserving grey eumelanosomes; (H) submalar feathers (sample G) preserving phaeomelanosomes. Yellow dots indicate location of each sample.

(Figs 2C and 3). The morphology of these bands is obscured by underlying body feathers, some of which can be clearly observed extending out from under the humerus (Fig. 3). The bands are not considered calami based on their interlocking orientation and proportions, being both too long and robust to be rachises of wing coverts. Similar bands are also preserved in the same anatomical position in *Eoconfuciusornis* IVPP V11977 (see Supplementary Fig. 5B, available as Supplementary Data at NSR

online); however, the preserved position of the wing in this specimen (folded towards the head so that the bands overlap with the dorsal neck feathers) obscures interpretations [25].

Although the partially contracted position of the wing in STM7-144 and overlap of the bands obscure the precise morphology of these internal structures, two overlapping tracts can be discerned, oriented at different angles. The clearest bands (tract A) extend along the dorsal margin of the humerus,

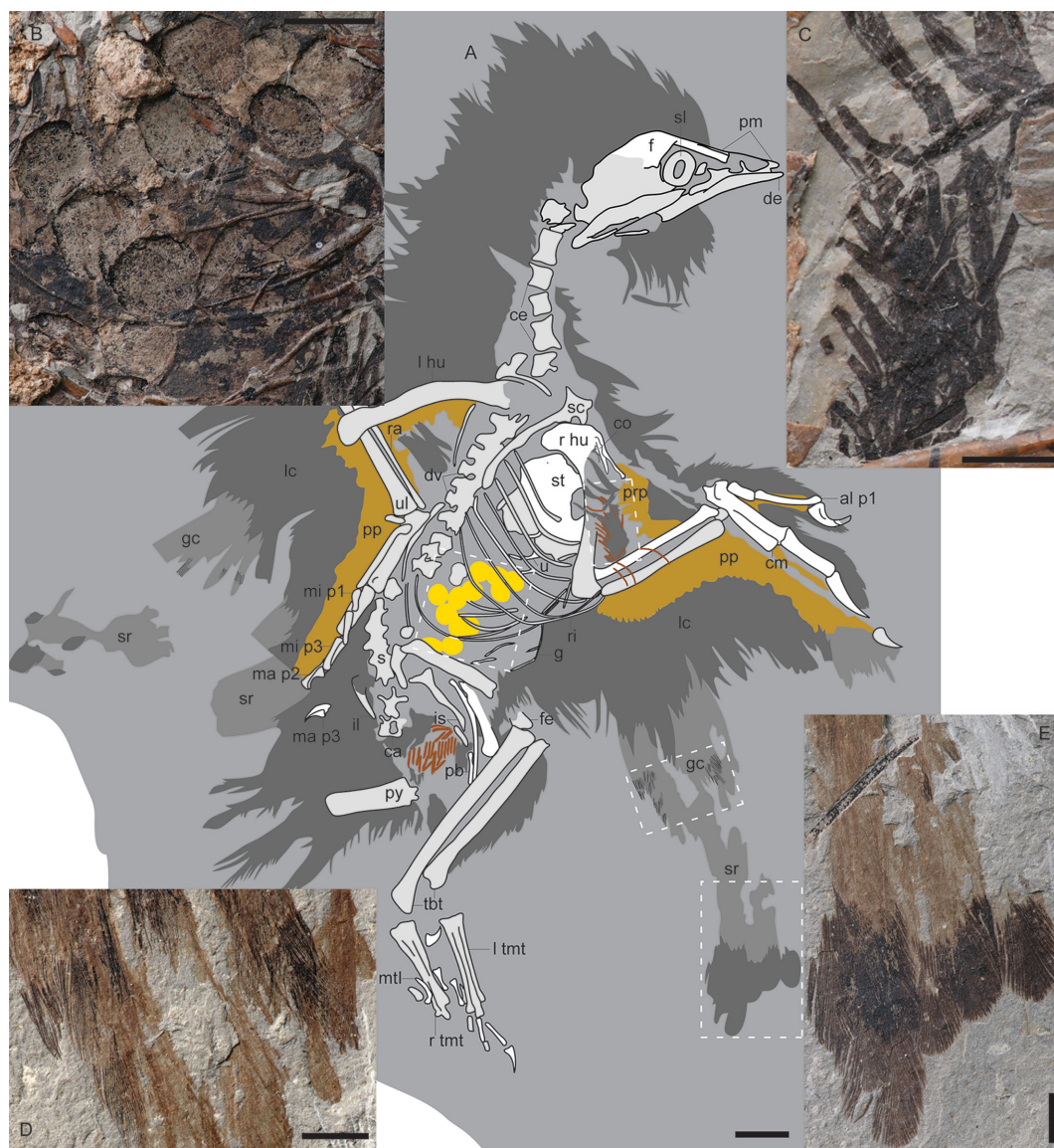


Figure 2. (A) Interpretative drawing of *Eoconfuciusornis* indet. STM7-144; (B) ovarian follicles; (C) remnants of the internal structure of the propatagium with underlying feathers; (D) preserved pattern in the greater coverts; (E) secondary remiges. Scale bar equals 10 mm in (A) and 5 mm in all insets. Anatomical abbreviations: al, alular metacarpal; ca, caudal vertebrae; ce, cervical vertebrae; cm, carpometacarpus; co, coracoid; cv, wing coverts; de, dentary; dv, dorsal vertebrae; f, frontal; fe, femur; g, gastralia; gc, greater coverts; hu, humerus; il, ilium; is, ischium; l, left; lc, lesser coverts; ma, major metacarpal; mc, marginal coverts; mi, minor metacarpal; mt, metatarsal; p1-3, manual phalanges; pb, pubis; pm, premaxilla; pp, postpatagium; prp, propatagium; py, pygostyle; ra, radius; ri, ribs; s, synsacrum; sc, scapula; sl, scleral ossicles; sr, secondary remiges; st, sternum; tbt, tibiotarsus; tmt, tarsometatarsus; ul, ulna; u, uncinat process. Yellow indicates preserved remnants of maturing ovarian follicles; dermal and epidermal tissue remnants are indicated in tan; remains of collagen fibres are brown. Dark grey indicates body feathers and the dorsal layer of wing coverts; light grey indicates the greater coverts and secondaries with preserved patterning. White dashed boxes represent areas enlarged in insets.

demarcating approximately a 60° angle with this bone (Fig. 3). The parallel bands are concentrated along the distal half of the humerus. The bands in tract A have an even thickness of 0.371 mm. Length is obscured by incomplete preservation and overlap with feathers. They appear to measure approximately 3.0–3.5 mm. A second tract (B) of overlap-

ping bands is also visible, demarcating a 30° angle with the humerus (Fig. 3).

These two ‘tracts’ of bands correspond in position and morphology to the two internal networks that support the propatagium in living birds [23]: tract A is consistent with remnants of the fascial curtain of the collagenous network, whereas tract

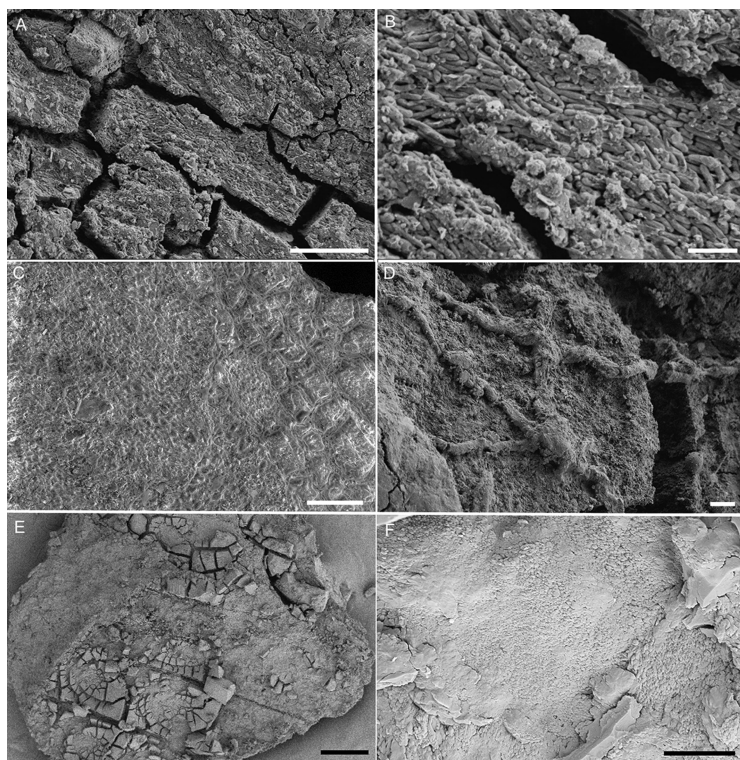


Figure 3. SEM images of samples from areas preserving the three types of soft tissue. (A) and (B) high-resolution images of the dorsal neck feathers (sample I) ((A), scale bar equals 10 μm ; (B), scale equals 2 μm); (C) and (D) propatagium ((C), sample F, scale bar equals 100 μm ; (D), sample 2.5, scale bar equals 10 μm); (E) and (F) follicle (sample A) ((C), scale bar equals 100 μm ; (D), scale bar equals 10 μm).

B appears to represent remnants of the fascial sheet of the elastic network (Fig. 3). As in STM7-144, in neornithines, these networks are visible to the naked eye [23]. The fascial sheet consists of elastic fibres extending from the lateral surface of the proximal end of the propatagial ligament near the proximal humerus and attaches to the dorsal layer of dermal tissue [23], consistent with the trajectory of the bands in tract B. The collagenous network consists of the fascial curtain and collagen web. The fascial curtain is primarily limited to the caudal half of the propatagium and consists of collagen fibres arranged into discrete thick and thin bundles that extend from the dorsal brachial fascia to the connective tissue surrounding the propatagial ligament. We suggest the traces (tract A) preserved in STM7-144 are the proximal ends of intact thick bundles of collagen fibres. The collagen web consists of interlacing semi-circles of collagen filaments extending from the brachial fascia to the periligament connective tissue and across the fascia of the antebrachium [23], traces of which are also preserved in STM7-144 (Fig. 3).

Plumage

STM7-144 is preserved with a nearly complete halo of carbonized feather remains (Fig. 1). These remains reveal the outline of the neck, propatagium and ventral surface of the body. In some areas, feathers are preserved overlying the bones whereas, in others, they are visibly underlying elements. Differences in the lengths of feathers covering different parts of the body clearly indicate differentiation into distinct pterylae (feather tracts), which is typically not apparent due to incomplete preservation of the plumage in most specimens. Body feathers in the dorsal tract are much longer than those in the ventral tract. No distinct apteria are visible due to overlap, although relatively little feathering is preserved on the ventral margin of the body. Within the capital tract, the submalar feathers are well developed, and dorsally the feathers elongate from the frontal to coronal region. Their rostral extent suggests the horny beak extended to the caudal margin of the external nares, as in living birds. The body feathers in the dorsal tract are preserved continuous with the body feathers in the caudal tract that covered the pygostyle. No elongate tail feathers are present. The crural feathers are short but well developed along the proximal third of the tibiotarsus. They are reduced distally, ending just proximal to the distal condyles. Well developed crural feathers reported in a specimen of *Confuciusornis* [26] are identified as overlapping traces of remiges (see Supplementary Fig. 6, available as Supplementary Data at NSR online).

Short feathers (marginal coverts) demarcate the leading edge of the propatagium (Figs 1 and 3A). These feathers project cranially; this is also observed in living birds and is important for creating the cambered profile of the propatagium [23]. Short feathers cover the cranial margin of the wrist and alular digit. At least one longer feather is preserved between the alular and major digits, extending the length of the alular digit (Fig. 3 and Supplementary Fig. 4A, available as Supplementary Data at NSR online). Short carpometacarpal coverts (covering the cranial edge of the hand) extend along the entire length of the major digit. The distal ends reveal short barbs neatly preserved in parallel, strongly suggesting that these feathers were pennaceous (Supplementary Fig. 4A, available as Supplementary Data at NSR online). Clearly visible in the right wing, a short layer of covert feathers is preserved projecting out of the postpatagium and extending up the caudal margin of the distal humerus (Figs 1 and 2). These and all body feathers are preserved black. A second layer of secondary coverts is also preserved but incomplete. These feathers are twice the length of the shorter coverts and clearly pennaceous (Fig. 2D).

Table 1. A list of samples taken from *Eoconfuciusornis* STM7-144 for SEM analysis. Feather samples indicate types of melanosome observed.

sample	substrate	melanosomes	color
a	follicles		
b	follicles		
c	follicles, matrix		
d	follicles, matrix		
rd	matrix		
A	remiges, dark	eumelanosomes	black
B	remiges, dark	eumelanosomes	black
C	remiges, light	eumelanosomes, phaeomelanosomes?	??
D	remiges, light	no results	
E	remiges, dark	no results	
F	patagium	eumelanosomes	gray
G	submalar	phaeomelanosomes	brown
H	dorsal neck	no results	
I	dorsal neck	eumelanosomes	black
J	dorsal tail	eumelanosomes	black
K	crural	eumelanosomes	gray
L	marginal	eumelanosomes	black
M	follicles		
N	ventral tail	no results	
2_1	coronal	eumelanosomes	gray
2_2	wing coverts	eumelanosomes	black
2_3	tail	eumelanosomes	black
2_4	crural	eumelanosomes	gray
2_5	patagium	eumelanosomes	gray

The remiges are the only incomplete portions of the plumage (Fig. 1). The distal portions of long pennaceous feathers are preserved below the greater coverts (Figs 1 and 2). These feathers are interpreted as secondary remiges based on their symmetrical vanes and rounded distal margins (Fig. 2E). In other confuciusornithiforms, including IVPP V11977, the primary remiges are asymmetrical and strongly tapered. The greater coverts and secondaries are preserved a brownish colour, with the exception of black spots that appear to represent remnants of the original patterning. The secondaries have dark tips and a pattern of dark spots is weakly visible on the greater coverts (Figs 1, 2D and E).

SEM analysis

We collected a total of 24 samples from STM7-144 for SEM analyses (Table 1); the primary focus was to investigate the internal structures of the propatagium, although the ovarian follicles, feathers and matrix were also sampled (see Supplementary Figs 7 and 8, available as Supplementary Data at NSR online). The propatagium sample reveals a lattice of

bands not observed in any of the other samples overlying a layer of melanosome-like mono-organelle structures similar to those in the feather samples (Fig. 4D). We suggest these bands are internal structural fibres of the propatagium overlying grey eumelanosomes from the ventral feathers (Fig. 2C). Energy-dispersive X-ray spectroscopy (EDS) of a follicle sample reveals primarily clay (the matrix) but also detects small amounts of calcium carbonate on the surface of the slab (see Supplementary Fig. 9B, available as Supplementary Data at NSR online).

Sixteen feather samples were collected: five specimens from the secondaries with three from the dark tips and the rest representing the major tracts of body feathers (Supplementary Fig. 1, available as Supplementary Data at NSR online). Samples from the dorsal neck, tail, marginal coverts and distal secondaries reveal sheets of carbon formed primarily by long (0.9–1.2- μm) and narrow ($\sim$ 0.2- μm , average aspect ratio 4.9) organized mono-organelles, which we interpret as eumelanosomes (Figs 1, 4A and B) [7,27]. This identification is supported by the preservation of beta-keratin [28]. Large oval eumelanosomes (1–1.3 μm long, 0.4–0.6 μm wide, average aspect ratio 2.3) were observed in the coronal and crural feathers (Fig. 1C and G). The submalar sample revealed small, fat phaeomelanosomes (0.4–0.6 μm long, 0.3 μm wide, average aspect ratio 1.7) (Fig. 1H). Small numbers of mouldic large grey eumelanosomes are also preserved in the light regions of the remiges (Fig. 1E) but not observed in the dark samples (Fig. 1D), which otherwise look the same under SEM (formed by sheets of carbon with superficial cracks produced during diagenesis) and have similar EDS profiles (see Supplementary Figs 9 and 10, available as Supplementary Data at NSR online).

DISCUSSION

The extraordinary preservation of soft tissues in *Eoconfuciusornis* STM7-144 sheds considerable light on the biology of the Confuciusornithiformes. The preservation of ovarian follicles allows STM7-144 to be confidently identified as a sexually mature female. Consistent with this fact, elongated tail feathers like those present in the only other specimen of *Eoconfuciusornis* IVPP V11977 [19] are absent. Although this absence could alternatively be due to moulting or incomplete preservation, we tentatively suggest that sexual dimorphism of the tail plumage was already present in the oldest and most basal recognized confuciusornithiforms (Fig. 5). Furthermore, despite its skeletal immaturity, IVPP V11977 is larger than STM7-144, strongly suggesting that sexual dimorphism in size also characterized



Figure 4. The propatagium of *Eoconfuciusornis* and neornithines compared. (A) Close-up of the right wing in STM7-144; (B) right wing with labels; (C) comparative illustration of neornithine propatagium (*Bubo virginianus* modified from Brown *et al.*, 1994). Yellow lines indicate the preserved traces of the fascial curtain (tract 'A') and collagen web, which together form the collagenous network; red indicates tract 'B', the fascial sheet of the elastic network. Differences in fibre orientation between STM7-144 and the illustrated neornithine result from differences in elbow extension. Dashed white lines indicate the borders of preserved dermal tissue. Anatomical abbreviations not listed in Fig. 2 caption: ial, incipient alula; cc, carpal coverts; ep, extensor process; hs, horny sheath; mc, marginal coverts; mp, metapatagium; pap, parapatagium; pl, propatagial ligament.



Figure 5. *In vivo* reconstruction of a male and female pair of *Eoconfuciusornis*. Artwork by Michael Rothman.

Eoconfuciusornis. The wings in STM7-144 preserve remnants of their original patterning, also observed in some specimens of *Sinosauropteryx* [29] and *Caudipteryx* [30] from the Jehol, considered indicative of *in vivo* colour disparity, indicating the wings were spotted. However, differences in the colour of preserved feathers in STM7-144 are only weakly supported by differences in melanosome morphology, and light and dark samples appear grossly similar under SEM. Similar *in vivo* colour patterns preserved in insects are also not reflected by differences under SEM or elemental analysis [31]. The taphonomy responsible for the preservation of these *in vivo* patterns is currently unknown; however, in fossil feathers, the differences may be structural [27]. Sampling for SEM analysis across the speci-

men is too sparse to fully reconstruct the plumage, but the predominance of eumelanosomes in most samples suggests that female *Eoconfuciusornis* were mostly black with a brownish throat patch, grey head and legs, and dark spotted wings (Fig. 5). Both eumelanosomes and phaeomelanosomes have been reported within a single feather of *Confuciusornis* also indicating complex patterning [4].

An alula (bastard wing) is a synapomorphy of Ornithothoraces, to which *Confuciusornithiformes* are commonly resolved as the out group [17,19]. In *Eoconfuciusornis*, at least one elongated pennaceous feather is attached to the proximal end of the alular digit representing an incipient stage in the evolution of the alula (Fig. 3). In enantiornithines and basal ornithuromorphs, the alular feathers attach more distally and extend nearly to the distal end of the major digit, as in living birds (Fig. 6). Pennaceous feathers are not otherwise known to be present on the alular digit of paravians, with the exception of the volant-dromaeosaurid *Microaptor* [32]. As in ornithothoracines, the alular feathers attach to the distal end of the digit (Fig. 6B); however, their distal extent is unknown [32].

Ovarian follicles

STM7-144 is the first *confuciusornithiform* and oldest known specimen to preserve ovarian follicles [11,19]. Due to the controversial nature of these findings [12,13], we revisit the strongest alternative that these are ingested items and, based on their shape, most likely represent seeds. In addition to anatomical location and morphology, three observations suggest this is not the case: first, these circular structures are proportionately large to be ingested seeds, being much larger than seeds preserved in birds of similar size [33]; second, among known specimens of Cretaceous birds, seeds have so far only been found preserved in the crop; and, third, the circular structures in STM7-144 are in some cases deformed into ovals, suggesting that they were not hard, like seeds (Fig. 2B and Supplementary Fig. 1, available as Supplementary Data at NSR online). Despite the fact that the number of collected *confuciusornithiform* specimens has been estimated at over 1000, none preserves stomach contents despite the fact that seeds are preserved in several specimens of *Jeholornis*, a taxon known from far fewer specimens [34,35]. Although seeds and gizzard stones are known in *Sapeornis* and several ornithuromorphs, no such evidence is present in *Confuciusornis* and Enantiornithes, interpreted as evidence of major dietary differences between these clades [36].

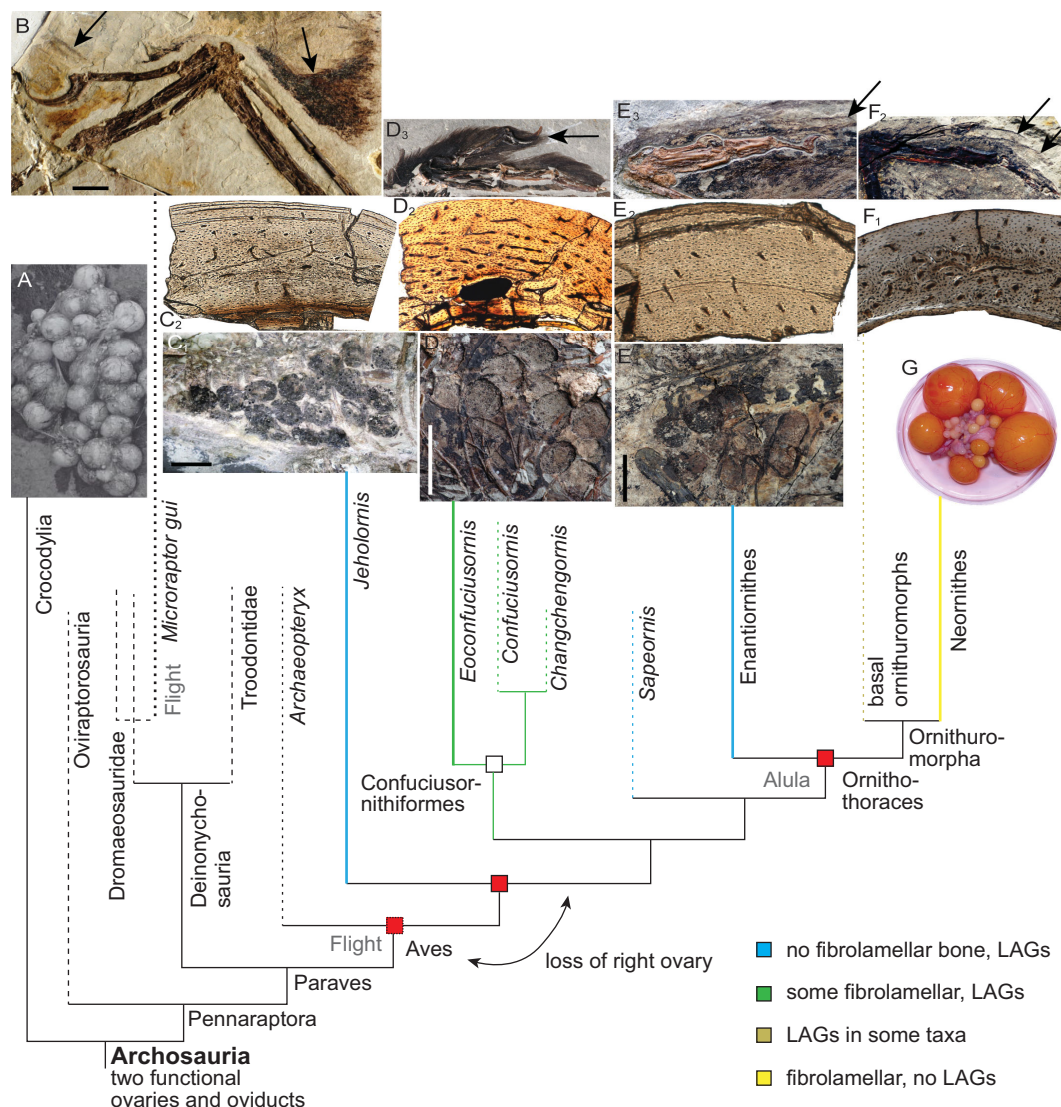


Figure 6. Simplified cladogram of Pennaraptora illustrating the distribution of the alula, propatagium and higher basal metabolic rates as inferred through bone histology and follicular morphology. Dashed lines refer to missing follicular data. (A) A single crocodilian ovary (paired *in vivo*) with a large number of similarly sized mature follicles; (B) *Microaptor gui* IVPP V13352, arrows indicate alula and propatagium; (C)₁₋₂, *Jeholornis* STM2-51 preserving ovarian follicles and histology of the ulna; (D)₁₋₃, *Confuciusornithiformes*: 1, follicles in STM7-144; 2, histology of humerus in unpublished subadult *Confuciusornis* STM13-40 showing fibrolamellar bone; 3, rudimentary alula (arrow) in STM7-144; (E)₁₋₃, *Enantiornithes*: 1, ovarian follicles preserved in STM29-8; 2, histology of the humerus of STM29-8; 3, alula preserved in *Eoenantiornis* IVPP V11537; (F)₁₋₂, *Ornithomorphs*: 1, histology of the ulna of *Iteavis* IVPP V18958; 2, alula (arrows) preserved in *Archaeornithura* STM7-145; (G) single left ovary of a domestic chicken *Gallus gallus*. Scale bar equals 10 mm.

Although STM7-144 is temporally older and phylogenetically regarded as stemward of the Enantiornithes [37], the preserved follicles show the greatest degree of follicular hierarchy observed so far in a Cretaceous bird (Fig. 6). The size difference between the largest and smallest preserved follicle is 2.3 mm compared to 0.4–1.7 mm in similarly sized enantiornithines [14]. Follicles preserved in *Jeholornis* show very little size disparity (minimal follicular hierarchy)—a condition similar to that in crocodil-

ians (Fig. 6), in which yolk is slowly deposited in all follicles simultaneously over several months, producing follicles of similar size [11,38,39]. Unlike crocodilians (and probably sauropod dinosaurs) who ovulate and ovideposit *en masse* so that yolk deposition and ovulation do not co-occur, birds (and potentially their closely related non-avian relatives) lay their clutches over a period of time, with individual ovulations separated by a period of at least 24 hours [40,41]. Neornithines also deposit yolk

over extended periods of time but experience a final rapid-growth phase that is not initiated in the entire clutch simultaneously, but rather in ovulatory sequence [20]. During this phase, greater amounts of yolk are assimilated as the follicle matures, facilitated by the increasingly larger network of blood vessels [20], producing a distinct hierarchy of pre-ovulatory vitellogenic follicles (Fig. 6G). This strategy serves to reduce the weight of the reproductively active ovary and reduce the necessary duration in which the ovary is swollen. Non-avian reptiles with a follicular hierarchy differ greatly from the avian condition, having two functional ovaries with active germinal beds (birds complete oögonial proliferation during embryogenesis) and a hierarchy that includes non-vitellogenic follicles or represents more than a single clutch [20,42]. The stronger follicular hierarchy observed in *Eoconfuciusornis* suggests that these birds evolved a more efficient reproductive system in parallel to neornithines. However, follicular hierarchy in STM7-144 is far less developed compared to living birds, suggesting that yolk deposition still occurred at a lower rate. This is consistent with the large number of developing follicles present in STM7-144; in neornithines with comparably large clutches, yolk deposition is so rapid that at no moment are all pre-ovulatory follicles developing concurrently [43]. The potential discovery of relatively faster yolk deposition in *Eoconfuciusornis* parallels histological data, which also indicates faster growth relative to other stem birds [44] (Fig. 6). Living birds have complex life-history energetic profiles, including large spikes in energetic demands during early post-natal growth and reproduction [45]. Data on the Confuciusornithiformes suggest that, physiologically, this lineage may have evolved similarly more complex energetic profiles in parallel to the neornithine lineage. The final rapid stage of yolk deposition unique to birds among living archosaurs increases energy efficiency and reduces the weight of the reproductively active ovary during yolk deposition and ovulation (which overlap in birds but not in crocodilians) [20].

Wing patagium

The avian propatagium is a fleshy aerodynamic surface that is responsible for the cambered profile of the wing extending from the proximal humerus to the wrist [23]—a feature absent in most flightless paleognaths [46]. This structure is preserved in several specimens of *Confuciusornis* [47], enantiornithines [9] and hongshanornithid ornithuromorphs (J. O'C. pers. obs.). In all these specimens, the wings are preserved partially folded,

obscuring the full surface area of the propatagium [9]. Typically, the propatagium is preserved as a faint, featureless impression (e.g. *Confuciusornis* IVPP V13156, *Hongshanornis* IVPP V14533) with the exception of MCCMLH31444, which preserves remains of the dermal and epidermal layers, revealing the distribution of feather follicles across the surface [9]. Contra [46,48], a propatagium is absent in *Archaeopteryx* and most non-avian maniraptorans (e.g. *Caudipteryx*) with the exception of the volant dromaeosaurid *Microraptor* (IVPP V13352; Fig. 5B) [46]. We suggest that, like other flight-related features (e.g. alula, hindwing) [49], a propatagium evolved independently in the microraptorine lineage. Living birds additionally have a smaller postpatagium along the caudal margin of the ulna and hand that covers the calami of the remiges and a small alular patagium that connects the alular and major digits [23], both of which have been recently documented in an enantiornithine [9]. *Eoconfuciusornis* preserves a postpatagium comparable to that in enantiornithines and living birds but an alular patagium is clearly absent (Fig. 3). The metapatagium and parapatagium of neornithines [23] are so far undocumented in Cretaceous birds.

Unlike the primary aerodynamic surface of the wing (formed entirely by feathers), the fleshy propatagium is formed by many layers (e.g. dorsal and ventral dermis, fibrous network, muscles, ligaments, fascia) and externally covered by feathers dorsally and ventrally [23]. In STM7-144, multiple layers of tissue are preserved, including feathers, some dermal tissue and what appear to be remnants of the collagenous and elastic support networks, consistent with the morphology, position, orientation and size of these structures in living birds [23] (Fig. 3). Traces of collagenous fibres are preserved in the postpatagium in MCCMLH31444 associated with the remige calami [9] and tentatively in an enantiornithine from the Early Cretaceous Xiagou Formation [25]. This specimen preserves similar bands near the pubis (see Supplementary Fig. 11, available as Supplementary Data at NSR online), where notably these bands are also preserved in STM7-144 (Figs 1 and 2 and Supplementary Fig. 5A, available as Supplementary Data at NSR online). This is not surprising considering that collagenous bands provide structural support for many parts of the body. The collagenous and elastic networks of the propatagium are responsible for supporting the cambered profile important for sustaining lift [23]. The presence of cranially projecting marginal coverts and an essentially modern support network strongly suggest the propatagium in this early pygostylian was well adapted to maintain

a cambered profile [23] and supplement the lift generated by the remiges [50].

METHODS

The specimen was observed under the binocular microscope and sampled manually for SEM analysis. Sample 'M' (see Supplementary Fig. 7, available as Supplementary Data at NSR online) and 'a–d' (see Supplementary Fig. 8, available as Supplementary Data at NSR online) are from the follicles (Table 1). We removed a piece of the slab where the follicles are embedded within the sediment, split the removed piece and took the upper half as our sample. The uncoated samples were then investigated under the scanning electron microscope (SEM Leo1530VP) with a Variable Pressure Secondary Electron (VPSE) Detector in a low-vacuum condition with a higher accelerating voltage (20 keV). EDS was used for selected point analysis of elemental compositions. All EDS data of the uncoated samples were collected at 20-keV accelerating voltage, at a working distance of 10 mm and for 120 seconds. A total of 16 feather samples were observed. The samples were removed at two different times. Images from the first group of samples (A–M) are low-resolution (high-resolution photographs taken from sample I) but these samples were later used in transmission electron microscopy tests and could not be re-imaged. The results of some of these samples (Table 1; D, E, H and N) are inconclusive. The second group of samples (2.1–2.5) were studied with a Secondary Electron (SE) detector in a high-vacuum condition with a lower accelerating voltage (3 keV), producing higher-resolution pictures. Due to limitations in permitted destructive sampling, these samples are unfortunately not from the critical regions and, because of the exceptional nature of this specimen (and singular nature of the slab), further destructive sampling is not possible.

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Conflict of interest statement. None declared.

SUPPORTING INFORMATION

Supplementary data are available at [NSRCP](#) online.

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